

GROUP AND RING THEORY - EXERCISE 14 (ANSWERS)

1. Find rational canonical forms of the following matrices over $\mathbb{Q}$:

$$(a) \begin{pmatrix} 1 & 5 & 7 \\ 0 & 4 & 3 \\ 0 & 0 & 1 \end{pmatrix};$$

$$(b) \begin{pmatrix} 2 & 4 & 0 \\ 1 & 4 & 1 \\ 3 & 8 & 3 \end{pmatrix}.$$

Solution. (a) We have

$$\begin{pmatrix} 1-x & 5 & 7 \\ 0 & 4-x & 3 \\ 0 & 0 & 1-x \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (x-1)^2(x-4) \end{pmatrix}$$

As $(x-1)^2(x-4) = x^3 - 6x^2 + 9x - 4$, the rational canonical form is $\begin{pmatrix} 0 & 0 & 4 \\ 1 & 0 & -9 \\ 0 & 1 & 6 \end{pmatrix}$.

(b) Here

$$\begin{pmatrix} 2-x & 4 & 0 \\ 1 & 4-x & 1 \\ 3 & 8 & 3-x \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & x^3 - 9x^2 + 14x - 8 \end{pmatrix}.$$

So the rational canonical form is $\begin{pmatrix} 0 & 0 & 8 \\ 1 & 0 & -14 \\ 0 & 1 & 9 \end{pmatrix}$.